

Supervisor Report: vec_both Optimization Findings

Executive Summary

Included cohort size: 44 subjects (q1>=65 with image score).

Best rm<=3 Spearman setting: removed_images='3;4;5', min_answers=4, n=41, r=0.311898, p=0.0471229, FDR=0.602686.

vec_both Calculation and Theory

Answer-level metric: sim_above_concept_total_vec_both, representing semantic alignment of each participant answer to the target concept using vector-based matching.

Subject-level image questionnaire score: arithmetic mean of sim_above_concept_total_vec_both over retained answers for that subject.

Formula: image_score_vec_both(subject) = mean(sim_above_concept_total_vec_both over retained answer rows).

This subject-level score is correlated with Questionnaire 1 score across subjects using Spearman rank correlation.

Methodology

Grid search: remove up to 3 images and sweep min_answers from 0 to 10.

Selection rule: maximize Spearman r (tie-break by smaller p).

Best vs No-Removal Baseline

case	removed_images	min_answers	n_subjects	r	p
best (rm<=3)	3;4;5	4	41	0.311898	0.0471229
best no-removal		0	44	0.202264	0.187938

Threshold Behavior for removed_images=3;4;5

min_answers	n_subjects	spearman_r	spearman_p
0	44	0.262236	0.0854976
1	44	0.262236	0.0854976
2	43	0.267548	0.0828281
3	42	0.251547	0.108061
4	41	0.311898	0.0471229
5	39	0.290024	0.073295
6	37	0.258116	0.122968
7	35	0.208147	0.230183